Lecture 28.12: Decision Tree Regressor

📌 Topic Overview

How does a Decision Tree work when the target variable is continuous (numerical) rather than categorical? In this lecture, we study the **Decision Tree Regressor**, understand how leaf predictions are calculated (mean/average), and examine the classroom numerical dataset.

🎯 Regression vs Classification in Trees

Property	Decision Tree Classifier	Decision Tree Regressor
Target Variable (y)	Discrete classes (e.g., Yes/No, Cat/Dog)	**Continuous numbers** (e.g., Salary, House Price, Age)
Leaf Prediction	Majority class label (mode)	**Mean / Average ($\bar{y}$)** of leaf samples
Impurity Metric	Entropy or Gini Impurity	**Variance (σ^2), MSE, or MAE**
Split Optimization	Maximize Information Gain	Maximize **Variance Reduction**

🍃 How Leaf Nodes Make Predictions: The Mean Value

In a classification tree, a leaf predicts the majority vote. In a regression tree, if a leaf receives samples with target values $\{y_1, y_2, \ldots, y_k\}$, the predicted value $\hat{y}$ is simply the **arithmetic mean**:

$$\hat{y} = \bar{y} = \frac{1}{k} \sum_{i=1}^{k} y_i$$

🧮 Blackboard Numerical Example: Target Dataset

In the video, the instructor writes a 4-sample dataset on the blackboard:

Feature (x)	Target (y)
1	2
2	4
3	6
4	10

1. Calculating the Mean ($\bar{y}$)

Total samples $n = 4$:

$$\bar{y} = \frac{2 + 4 + 6 + 10}{4} = \frac{22}{4} = 5.5$$

2. Calculating the Parent Node Variance (V_S)

The variance formula is:

$$V_S = \text{Var}(y) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Calculating squared deviations from the mean ($\bar{y} = 5.5$):

- $(2 - 5.5)^2 = (-3.5)^2 = 12.25$
- $(4 - 5.5)^2 = (-1.5)^2 = 2.25$
- $(6 - 5.5)^2 = (0.5)^2 = 0.25$
- $(10 - 5.5)^2 = (4.5)^2 = 20.25$

$$\text{Sum of Squared Errors} = 12.25 + 2.25 + 0.25 + 20.25 = 35.0$$

$$V_S = \frac{35.0}{4} = 8.75$$

The instructor highlights on the blackboard: $V_S = 8.75$! This is the initial impurity of the parent node.

💻 Python Implementation

```
import numpy as np

# Classroom dataset
x = np.array([1, 2, 3, 4])
y = np.array([2, 4, 6, 10])

# Mean prediction
mean_y = np.mean(y)

# Variance calculation
variance_y = np.var(y)

print(f"Classroom Target Values: {y}")
print(f"Parent Mean (y_bar):    {mean_y:.2f}")
print(f"Parent Variance (V_S):   {variance_y:.2f}")
assert np.isclose(variance_y, 8.75), "Variance must match blackboard calculation (
```

🔑 Key Takeaways

1. Decision Tree Regressors predict continuous numbers.
2. A leaf node's prediction is the **mean (average)** of the target values in that leaf.
3. Node impurity is measured using **Variance** or **Mean Squared Error (MSE)** instead of Entropy.